Question	Answer	Marks	Partial Marks
8(a)	6 nfww	**3**	**M2** for $\frac{2.65-2.50}{2.50}[\times 100]$ or for $\frac{2.65}{2.50}\times 100$ or **M1** for $\frac{2.65}{2.50}$
8(b)	552.5[0]	**3**	**B2** for 52.5[0] or **M2** for $500\times\frac{1.5}{100}\times 7+500$ oe or **M1** for $500\times\frac{1.5}{100}$ $[\times 7]$ oe
8(c)	37.4 or 37.36…	**2**	**M1** for $\left(1+\frac{1.6}{100}\right)^{20}$ oe soi 1.37…
8(d)	4[.00...]	**3**	**M2** for $\sqrt[22]{\frac{2607}{6400}}$ or **M1** for $6400\times x^{22}=2607$ oe or better
9(a)	82	**2**	**M1** for $(3^x)^2+1$ soi by $(3^2)^2+1$ or g(9) isw
9(b)	$\frac{x+2}{7}$ final answer	**2**	**M1** for $y+2=7x$ or $\frac{y}{7}=x-\frac{2}{7}$ or $x=7y-2$
9(c)	[a =] 1, [b =] 2, [c =] 2	**3**	**B2** for $x^4+x^2+x^2+1+1$ or **M1** for $(x^2+1)^2+1$
9(d)	$\frac{6}{7}$ oe	**3**	**M2** for $7x-2=4$ or **M1** for $3^x=81$ soi f(x) = 4 or for $3^{7x-2}=81$ or better
10(a)	10	**1**	
10(b)	6.2[0] or 6.203 to 6.204	**3**	**M2** for [x^3 =] $1000\div\frac{4}{3}\pi$ oe or better or **M1** for $\frac{4}{3}\pi x^3=1000$
10(c)	7.82 or 7.815 to 7.816	**4**	**B3** for [x^3 =]$1000\div\frac{1}{3}\pi\div 2$ oe or better or **M1** for $\left(x\sqrt{5}\right)^2-x^2$ soi by $4x^2$ or $2x$ **M1dep** for $\frac{1}{3}\pi\times x^2\times theirh[=1000]$

Question	Answer	Marks	Partial Marks
10(d)	$6\frac{2}{3}$ or 6.67 or 6.666 to 6.667	**4**	**B3** for $[x^3 =]1000 \div \frac{27}{8}$ oe or $\frac{3x}{2} = 10$ or better or **M2** for $\frac{1}{2} \times x \times \frac{x}{2} \times \frac{27x}{2} = 1000$ oe or **M1** for $\frac{1}{2} \times x \times \frac{x}{2}$ If 0 scored, **SC2** for answer 5.29 or 5.291..
11	[Total time =]16 h 6 min or 16.1 h	**2**	**B1** for 22 h 6 min or 22.1h or 966 mins If 0 scored, **SC1** for 9 h 41 min
	[Distance to airport in New York =] 16.5	**2**	**M1** for 18×55
	[Arc length =] 6200 or 6199 to 6200. …	**3**	**M2** for $\frac{55.5}{360} \times 2 \times \pi \times 6400$ or **M1** for $\frac{55.5}{360}$ or $2 \times \pi \times 2400$
	[Distance Geneva to Chamonix =] 104	**2**	**M1** for 65×1.6 or 65×96 oe
	392 to 393	**2**	**M1** for $\frac{6316 \text{ to } 6322.4}{\textit{their}\,16.1}$ Must be correct value in numerator